

ASTON UNIVERSITY

EP4DIS MSc DISSERTATION

*DEVELOPMENT AND EVALUATION OF AN ARTIFICIAL INTELLIGENCE-
ENHANCED MOBILE PLATFORM FOR SUPPORTING NICOTINE VAPING
CESSATION IN THE UNITED KINGDOM*

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ABSTRACT

This report documents the design, evidence base, and implementation of a minimum viable product for VapeFree, a free mobile application intended to help people in the United Kingdom stop vaping nicotine. Nicotine vaping has grown rapidly in recent years, yet dedicated digital support for cessation remains limited, and the applications that do exist are commonly held back by subscription paywalls, rigid quitting methods, and technical unreliability. VapeFree responds to this gap with a product that is free at the point of use, that supports gradual reduction as well as abrupt cessation, and that is designed around the way people actually vape.

The work follows a pragmatic mixed-methods approach. Two research strands were carried out to ground the design in evidence: an automated analysis of 1,178 user reviews across nine competitor applications, and a primary survey of prospective users. The competitor analysis pipeline was implemented in Python with data governance built into its design, and it passed a full suite of automated tests. The evidence from both strands converges. Users want gradual reduction, they reject paywalls, they value simple and reliable tracking, and they abandon applications that are complicated, buggy, or that erase their progress. These findings were translated into a concrete feature design and a working minimum viable product, which was subjected to technical validation as reported in Chapter Four. Usability testing with participants and a pilot study remain planned as the next evaluation phases; they are not reported here because recruitment and data collection are still outstanding.

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CHAPTER ONE - INTRODUCTION

What the Project Is

VapeFree is a cross-platform mobile health application designed to help people in the United Kingdom stop vaping nicotine. It was developed as a minimum viable product using a lean, evidence-based approach. Its defining principles are that cessation support should be free at the point of use, flexible enough to accommodate different quitting styles, and built specifically around vaping behaviour rather than around habits carried over from cigarette smoking.

The project brings together three disciplines. It applies software engineering through cross-platform development and cloud data architecture, it applies artificial intelligence through the integration of a large language model for conversational support, and it applies public health through the use of established behaviour change methodology and human-centred design. The intended outcome is a working application, together with evidence of its feasibility, user acceptance, and preliminary effectiveness.

Why the Project Is Important

The United Kingdom is facing a significant public health challenge. Nicotine vaping prevalence has increased roughly sixfold over the past thirteen years, and current estimates indicate that around 5.5 million adults regularly use vaping products (Hartmann-Boyce et al., 2022; McNeill et al., 2023). Uptake is especially high among young adults aged 18 to 24, where prevalence reaches 26.5 per cent (Action on Smoking and Health, 2024). Of particular concern, approximately one million people who never previously smoked conventional cigarettes now vape regularly (Public Health England, 2023), which suggests that vaping is establishing new patterns of nicotine dependency beyond its intended role in harm reduction.

Despite the scale of the problem, dedicated digital support is scarce. National Health Service provision consists largely of general web guidance and in-person counselling with inconsistent

expertise in vaping. Commercial applications, meanwhile, present recurring barriers: annual subscription fees of roughly 84 to 155 pounds, designs that support only abrupt cessation, and poor fit to the disposable devices that now dominate the UK market. This combination leaves a large and growing population without accessible, evidence-based, and affordable help. Addressing that gap is both a public health need and an opportunity for digital health innovation.

Aim and Objectives

The project aims to design, develop, and evaluate a mobile health intervention that delivers cost-free, evidence-based cessation support for people who want to stop vaping nicotine, with the longer-term ambition of a design suitable for National Health Service integration. This aim is pursued through five objectives:

- **Requirements analysis and design:** analyse existing applications and conduct primary research to establish what users need.
- **Application development:** build a cross-platform application with behavioural tracking, a gradual reduction mode, an AI-powered chatbot, and a cloud backend.
- **Validation and testing:** carry out technical validation, accessibility evaluation, and usability testing.
- **Pilot study:** recruit participants, monitor engagement and outcomes, and conduct qualitative interviews.
- **Integrated analysis:** combine quantitative and qualitative findings to explain engagement and guide future development.

CHAPTER TWO - RELATED WORK

This chapter reviews the work that VapeFree builds upon: the research literature on digital cessation, the existing applications that occupy the market, and the techniques and technologies available for building an intervention of this kind. It closes by identifying the gap that the project addresses.

Digital Health Interventions and Behaviour Change

Mobile health interventions have a strong evidence base for smoking cessation. Systematic reviews indicate that smartphone applications can increase long-term abstinence by between 50 and 67 per cent compared with minimal interventions (Whittaker et al., 2019). The mechanisms behind this include accessibility, reduced stigma, personalisation, and scalability. Evidence specific to vaping cessation is far thinner. While over four hundred smoking applications exist, fewer than ten target vaping, and most simply adapt smoking content without accounting for the different devices, usage frequency, and demographics involved.

Effective interventions draw on established behaviour change techniques. At a broader level, the COM-B model (Michie et al., 2011) frames behaviour as the product of capability, opportunity, and motivation, and provides a structuring framework for identifying which techniques to deploy and when. The taxonomy developed by Michie et al. (2013) identifies techniques with robust support, including self-monitoring, goal setting, behavioural feedback, social support, and relapse prevention. Self-monitoring in particular is associated with higher engagement and better outcomes, although designs must balance thoroughness against user burden, since excessive complexity predicts abandonment. Cognitive Behavioural Therapy provides a complementary foundation, being the established basis for addiction treatment, with digital delivery shown to be broadly non-inferior to face-to-face delivery (Lancaster and Stead,

2017). Self-Determination Theory further suggests that interventions supporting autonomy, competence, and relatedness achieve stronger engagement (Ryan and Deci, 2000).

Existing Vaping Cessation Applications

A set of nine vaping cessation and tracking applications currently available on the Google Play Store was identified as the closest existing work. These products share a common core of features, typically a counter for days without vaping, a display of money saved, and in some cases a community feed or a puff-counting tool. They also share recurring weaknesses that this project examined empirically, with the method described in Chapter Three and the detailed results reported in Chapter Four.

At a high level, the landscape divides clearly. The better-regarded products are simple, focused, and reliable trackers. The poorly regarded products are undermined by aggressive subscription paywalls, by technical faults such as crashes and data loss on update, and by rigid tracking that resets a user's progress to zero after a single slip. Community features are valued when present but are frequently weakened by a lack of moderation. This landscape defines both the standard VapeFree must meet and the specific failures it must avoid.

Techniques and Technologies

Several established techniques and technologies underpin the build. On the application side, cross-platform frameworks such as React Native with Expo allow a single codebase to serve both iOS and Android, reducing development effort. Backend platforms such as Firebase provide managed data storage, offline synchronisation, and analytics without the overhead of self-hosting. Offline-first architecture, in which the application functions without a network connection and synchronises later, is well suited to a tracking tool that users open frequently and briefly.

On the artificial intelligence side, contemporary large language models offer natural, contextually appropriate conversation that earlier rule-based systems could not (Laranjo et al., 2018). Their use in health contexts is promising but carries known risks, including factual inaccuracy and inappropriate responses, so safety mechanisms such as crisis-language detection and medical-advice boundaries are considered essential. On the research side, automated review collection combined with rating-based sentiment classification and keyword-based theme extraction provides a scalable way to learn from large volumes of user feedback, while pseudonymisation techniques such as salted cryptographic hashing allow this to be done in a privacy-preserving manner. Accessibility is governed by the WCAG 2.1 standard, which specifies contrast, touch target, screen-reader, and plain-language requirements.

The Gap Addressed

Taken together, the literature and the existing tools reveal four gaps. First, there is a shortage of vaping-specific interventions that account for vaping behaviour rather than adapting smoking content. Second, no widely available application properly supports structured gradual reduction, despite clear user demand. Third, modern large language models have not been evaluated within accessible cessation support with appropriate safety frameworks. Fourth, the most effective features are commonly locked behind paywalls that exclude the users who most need help. VapeFree is positioned to address all four.

CHAPTER THREE - METHODOLOGY

Overall Approach

The project uses a pragmatic mixed-methods design combined with a lean product development philosophy. Rather than building a complete application before testing any assumptions, research informs design, design informs the build, and the build is validated with users before wider release. Both quantitative and qualitative evidence are used throughout: quantitative data provides an objective picture of behaviour and outcomes, while qualitative data explains why those patterns occur. The work is organised into five sequential phases spanning roughly twenty-two weeks, covering requirements elicitation, software development, usability evaluation, a pilot study, and integrated analysis. Feature decisions are grounded in the behaviour change frameworks reviewed in Chapter Two, principally the COM-B model and the Behaviour Change Technique Taxonomy (Michie et al., 2011; Michie et al., 2013).

Design

The design of the application was derived from two sources of evidence: a systematic analysis of existing tools through their user reviews, and a primary survey of prospective users. This section describes how each was carried out and how the final design follows from them.

Competitor Review Analysis

To learn systematically from existing tools, an automated command-line pipeline was implemented in Python. It connects to the Google Play Store through the google-play-scraper package and downloads reviews for nine pre-defined vaping cessation applications. In total, 1,178 reviews were collected. For each review, the pipeline captured a unique identifier, the star rating from one to five, the review text, the date, the count of helpful votes, and any developer reply. The data was parsed, cleaned, and stored in a relational SQLite database organised into tables for governance records, application details, reviews, and an audit log.

Compliance was built into the pipeline from the outset. Raw usernames are never stored. Each name is stripped, lowercased, combined with a secret cryptographically random salt held only in a local environment file, and hashed using SHA-256 to produce a deterministic sixteen-character identifier. Processing takes place under UK GDPR Article 89, the research exemption, with a strict twelve-month retention limit enforced by an automated expiry routine, and every significant action is written to an immutable audit log. Sentiment was then classified from the star rating, treating four or five stars as positive, three as neutral, and one or two as negative, and four themes were extracted through keyword matching: Paywalls and Cost, Tracking and Statistics, Community and Support, and Bugs and Account Setup. The pipeline itself was validated with a suite of automated unit and integration tests, all eight of which passed.

Competitor Analysis Findings

Average ratings varied widely across the nine applications, as shown below. The pattern is instructive: the highest-rated products are simple, focused trackers, while the lowest-rated are undermined by crashes and data loss.

Application	Average rating	Note
Unpuff	5.00	Small review base
Quit Addiction	4.71	
Escape the Vape	4.16	Solid, well-rated
Puff Count	4.00	
VapeNot	4.00	
Quit Buddy	3.53	Largest review base (N = 729)
VapeAway	3.00	
Quit Vaping	1.36	Bugs wipe out user data

Escape the Vape holds a solid rating of 4.16, while Quit Buddy, which carries the largest share of reviews at 729, sits at only 3.53 due to crashes, widget bugs, and rigid progress trackers.

Quit Vaping records a severely low rating of 1.36, driven almost entirely by bugs that erase user data. The clear message is that technical reliability, and the protection of user progress in particular, is decisive for satisfaction.

Theme analysis reinforces this. Paywalls and Cost, together with Bugs and Account Setup, are heavily associated with negative reviews, whereas Tracking and Statistics and Community and Support are the strongest drivers of positive reviews. The split between what earns praise and what causes complaints is summarised below.

Theme	What users praise	What users complain about
Paywalls and Cost	A genuine free tier.	Aggressive subscriptions and locked milestone pages.
Tracking and Statistics	Days clean and money saved counters.	Inaccurate calculations, rigid resets, and no reduction option.
Community and Support	Peer stories, mutual support, and solidarity.	Weak moderation, no direct replies, and toxic posts.
Technical Quality	Fast startup, minimal interface, responsiveness.	Crashes, data loss on update, and missing widgets.

User Needs Survey

Alongside the competitor analysis, a survey was designed to capture the preferences of prospective users directly. Ethics approval for the survey was obtained from Aston University prior to data collection. The instrument covers vaping history and patterns, previous quit attempts and methods, motivation to quit, preferred quitting approach, desired features, attitudes toward an AI chatbot, the importance of the application being free, and likely barriers to use, together with two open questions for further suggestions. The responses analysed in this report represent the initial batch of four ($n = 4$) collected during early requirements gathering. They are indicative rather than statistically representative, and the sample is expected to grow toward the target of fifty to one hundred participants set out in the project plan. Their value at this stage is that they are consistent with, and reinforced by, the much larger competitor review corpus.

Survey Findings

The initial survey respondents (n = 4) fall within the core target demographic, all aged between 18 and 34, spanning students and people in full-time employment, and using disposable products, pod systems, and vape pens or mods. Although small, the sample covers the main device categories a UK-focused application must support. Several consistent signals emerge, each aligning with the competitor analysis:

- **A clear preference for gradual reduction:** every respondent selected gradually reducing usage over several weeks as their preferred in-app approach, which directly supports making gradual reduction central to the design.
- **High cost sensitivity:** most respondents rated a completely free application as very or extremely important, with one rating it moderately, and several named subscription costs among the factors that would stop them using a quitting application.
- **Value in tracking and information:** commonly requested features included craving logs, money saved calculators, progress tracking of vape-free days, gradual reduction plans with daily targets, and health improvement milestones.
- **Mixed attitudes to an AI chatbot:** responses ranged from cautious interest to clear reluctance, which supports offering conversational help as an optional aid rather than a central feature.
- **Complexity as a barrier:** beyond cost, respondents cited applications being too complicated and not proving helpful as reasons to stop using them, reinforcing the case for a simple, immediately useful design.

Final Design Derived from the Evidence

Both strands point firmly in the same direction, which gives confidence that the design rests on evidence rather than assumption. The two strands were combined into a feature design organised as four decisions: what to add, what to improve, what to avoid, and what to retain. The single most important decision, supported by both strands, is that VapeFree will launch as a fully free tool, directly resolving the most common complaint about existing applications. The table below records the final design and traces each decision to its evidence.

Design decision	Description	Evidence
Gradual reduction mode	Puff logging, daily limits, and a tapering target	Survey (all respondents)

Design decision	Description	Evidence
	plan, offered alongside abrupt cessation.	preferred gradual); competitor gap.
Non-judgemental slip-up log	Records slips while preserving cumulative progress rather than resetting to zero.	Competitor complaints about punitive resets.
Offline-first anonymous use	Tracking starts immediately on download with no mandatory login.	Competitor complaints; privacy concerns.
Fully free, no ads	No paywalls, subscriptions, or intrusive advertising.	Both strands; strongest single signal.
Reliable tracking and backups	Simple counters with robust sync so updates never lose data.	Competitor data loss; survey feature requests.
Optional AI support	Conversational help offered as an opt-in aid, not a mandatory feature.	Mixed survey attitudes to a chatbot.

Implementation

The implementation of VapeFree followed the lean, evidence-based plan established in the design chapter. The mobile client was built as a managed Expo application using React Native, TypeScript, and Expo Router, with NativeWind providing Tailwind-style styling aligned to the Google Stitch wireframes. Firebase provides the backend: Authentication for anonymous user sessions, Firestore for structured behavioural data with a persistent local cache for offline-first reads and writes, and Cloud Functions in the europe-west2 region to proxy requests to Google Gemini. The Gemini API key is stored only on the server; the client communicates with the model exclusively through HTTPS callable functions.

Core product behaviour is organised around five primary tabs (Home, Track, Support, Progress, and Journal), together with an adaptive onboarding flow that supports both gradual reduction and abrupt cessation. The gradual reduction path applies a documented tapering algorithm with configurable pace and minimum daily limits; puff and craving events are logged on the Track screen and synchronised to Firestore when connectivity is available. Progress milestones, including streak and money-saved indicators, are derived from logged sessions rather than punitive reset logic. When a user logs a slip, a supportive reframing screen reinforces continuity of progress rather than treating a single event as total failure.

The Support tab delivers optional conversational help. When online, messages are handled by a Cloud Function that wraps Gemini with a fixed system prompt, crisis and medical-advice detection, and append-only chat logging suitable for researcher review. Users can report inappropriate assistant responses through a server-side callable function. When offline or when Functions are unavailable, a rule-based offline support companion matches user text to curated, safety-filtered responses without calling any external model. Analytics hooks record key product events in development and on web when Firebase Analytics is configured.

Accessibility was treated as a cross-cutting requirement rather than a late add-on. Shared text components respect system font scaling, interactive elements carry accessibility roles and labels, and `eslint-plugin-react-native-a11y` supports ongoing review. Pseudonymised research identifiers are assigned through a dedicated Cloud Function mapping participant codes to authentication identifiers, keeping exportable behavioural data separable from direct identity in line with the project's GDPR-oriented approach.

During iterative development and formative technical evaluation, the application was run on physical hardware through Expo Go with Metro bundler, connected to the Firebase Emulator Suite on the development host. This avoided production cloud cost while exercising authentication, Firestore sync, and the Functions proxy locally. A standalone Android APK was also produced through EAS Build for optional distribution experiments; the main evaluation path remained Expo Go with emulators, as documented in Chapter Four. Full production deployment remain outside the minimum viable product delivered at this stage.

CHAPTER FOUR - TESTING AND RESULTS

Introduction

This chapter documents how VapeFree was tested during development and how those tests support the evaluation approach described in Chapter Five. The application targets iOS and Android via React Native and Expo. For technical validation and formative usability work, the primary configuration was Expo Go with Metro (npm start) and the Firebase Emulator Suite running on the developer machine. This setup connected the mobile client to Authentication, Firestore, and Cloud Functions (including the Gemini proxy) without requiring a production deployment or paid cloud usage during iterative testing.

The chapter first outlines the test strategy, then describes the test environment and backend connection, reports the automated test results, summarises the manual and device-based testing, and previews the usability evaluation outcomes, which are cross-referenced to Chapter Five.

Test strategy

Testing followed a risk-based test pyramid appropriate for a single-developer MSc prototype:

- **Unit tests:** pure business logic in the lib/ layer (reduction algorithm, streaks, milestones, offline support matching, chat safety filters, and journal helpers). These tests are fast, repeatable, and do not depend on network or Firebase.
- **Component tests:** React Native Testing Library tests for critical UI (support chat bubbles, report flow, offline banner, milestone cards, and branding), including accessibility labels aligned with the WCAG 2.1-oriented design goals.
- **Smoke end-to-end flows:** Maestro scripts for tab navigation and the support screen, executed manually against Expo Go when Metro and the emulators were running.
- **Manual device testing:** day-to-day verification on a physical phone via Expo Go, with the Firebase emulators on the host machine.

Automated tests prioritised behaviour-change logic (gradual reduction caps and daily progress) and safety-critical chat behaviour (crisis and medical-advice detection, offline companion responses, and user reporting of assistant messages). Firebase client wiring and live Gemini

responses were validated primarily through emulator-backed manual sessions rather than full automated integration tests against production APIs.

Test environment and connection to Firebase

The client never calls the Gemini API directly; chat requests use HTTPS callable Cloud Functions. For development and testing, environment variables in a local .env file (not committed to version control) configured the app to use emulators when EXPO_PUBLIC_USE_FIREBASE_EMULATORS was set to true.

The evaluation configuration was as follows:

- **Mobile client:** Expo Go, loading JavaScript from Metro (npm start with LAN enabled).
- **Backend:** Firebase Emulator Suite on the developer's Mac (npm run emulators, or the combined npm run dev), exposing Auth (port 9099), Firestore (8080), and Functions (5001).
- **Physical device connectivity:** EXPO_PUBLIC_FIREBASE_EMULATOR_HOST set to the Mac's LAN IP so the phone on the same Wi-Fi could reach the emulators (localhost on the device would incorrectly refer to the phone itself).
- **Functions secrets:** GEMINI_API_KEY supplied via functions/.env for local function emulation only.

This arrangement was chosen because it simplifies backend connectivity during testing compared with standalone builds, keeps pseudonymised research data under local control during early trials, and matches the stack defined in the project proposal (Firebase Auth, Firestore, and Cloud Functions). Production Firebase web keys and EAS-hosted environment variables were reserved for optional deployment scenarios (for example, an internal testing APK) and were not required for the main Expo Go evaluation path.

Static quality gates used alongside the tests included TypeScript strict checking (npm run typecheck) and ESLint (npm run lint).

Automated testing: implementation and results

Tooling: Jest 29 with the jest-expo preset and @testing-library/react-native. Tests reside in lib/__tests__/ and components/**/__tests__/. Jest loads .env during setup so that emulator-configuration tests reflect local defaults, and AsyncStorage and Firebase Analytics dependencies are mocked where needed.

The scope of automated testing is summarised below:

Area	Modules / components tested	Rationale
Reduction plan	reduction-algorithm, reduction-pace, plan-profile, daily-plan-progress	Core adaptive cessation logic
Progress	streak-calculator, milestones, milestone-progress, chart helpers	Motivation and feedback accuracy
Journal	build-journal-entry, journal-mutations	Data shape before Firestore sync
Chat safety and offline support	chat-safety, offline-support-matcher, merge-chat-messages, offline-chat-repository	Ethical and offline requirements
Reporting and UI	ChatMessageBubble, offline banner, MilestoneCard	Research review and accessibility
Analytics	analytics-service	No crash when Analytics is unavailable on native
Firebase config	Emulator demo config when emulators enabled	Reproducible local setup

The results of the final development run were as follows:

Metric	Outcome
Test suites	25 passed, 0 failed
Test cases	68 passed, 0 failed
Execution time	Approximately 8 to 13 seconds on development hardware
TypeScript	Strict check passed (tsc --noEmit)

Coverage: Jest coverage was collected for lib/ and components/. Overall line coverage was approximately 39 per cent, with near-complete coverage on high-risk pure modules (for example, the reduction algorithm, streak calculator, offline chat repository, and message-merge logic) and lower coverage on the thin Firebase repository wrappers and hooks, which were

exercised manually via the emulators. This reflects a deliberate prioritisation of algorithmic and safety logic over exhaustive UI coverage within the scope of an MSc project.

Smoke end-to-end: Maestro flows (smoke-navigation and chat-screen) were defined under `.maestro/flows/` for Expo Go. They were used as optional smoke checks rather than continuous-integration gates, and their results depended on Metro and emulator availability at run time.

Manual and device-based testing results

Manual testing on a physical Android device via Expo Go confirmed the following:

- Application launch, tab navigation (Home, Track, Support, Progress, and Journal), and font loading.
- Anonymous authentication and Firestore read/write against the emulated backend (verified in the Emulator UI at <http://localhost:4000>).
- Craving and puff logging with offline-first behaviour when connectivity was interrupted (local cache, with synchronisation once online).
- Support chat via the Cloud Function proxy when the Functions emulator and Gemini key were configured.
- Offline support-companion responses when the network was unavailable (a curated local matcher, with no model call).
- The report action on assistant messages (server-side reporting via a callable function in the emulator context).

The following operational constraints were documented during testing:

- Expo Go requires an active Metro connection on the same network; disabling Wi-Fi produces Metro disconnect errors, which is expected behaviour for development builds rather than a functional defect of the application logic.
- The Firebase emulators require Java and must be running before backend-dependent features are tested.
- A standalone Android APK was built successfully via EAS Build for optional distribution experiments; the primary evaluation path remained Expo Go with emulators because the Firebase environment variables were not baked into that preview build.

Usability evaluation and outcomes ([link to Chapter Five](#))

Technical testing established that the platform is functionally stable under the Expo Go and emulator configuration, and that the automated tests pass for the core algorithms and safety-related behaviour. Whether the application is easy and helpful for real users will be assessed through the usability testing and pilot study planned in Chapter Five (for example, structured tasks with participants and the System Usability Scale). Those user-study results are reported in Chapter Five rather than in this technical chapter.

Chapter summary

VapeFree was tested using 68 automated unit and component tests (all passing), optional Maestro smoke flows, and systematic manual testing on Expo Go connected to the Firebase emulators on the host machine. This configuration was selected to simplify backend connectivity and to support iterative dissertation development. The technical results support proceeding to the usability evaluation in Chapter Five; the summative user outcomes will complete the "results" narrative of this chapter once participant testing is finished.

Testing setup summary

The key additions made to enable testing are summarised below:

Item	Purpose
.env with emulator flags and LAN host	Routes the app to the Mac emulators from the phone
functions/.env with GEMINI_API_KEY	Enables chat in the Functions emulator
Java runtime	Runs the Firestore, Auth, and Functions emulators
Expo Go with npm start	Client delivery without a standalone build
Jest and React Native Testing Library (68 tests)	Regression on logic and critical UI
Maestro flows	Optional navigation smoke tests

CHAPTER FIVE - FUTURE WORK

This chapter sets out the planned next phases of the project and a set of further ideas for improving the platform. The immediate priorities are to strengthen the evidence base, complete the application, and evaluate it with real users.

Expanded Survey Recruitment

The survey findings reported here are based on an initial batch of responses and should be treated as indicative. The next step is to grow the sample toward the planned target of fifty to one hundred participants and to broaden it across age, gender, and socioeconomic background so that the findings are more representative. Semi-structured interviews with a smaller group are also planned, in order to add depth to the quantitative preferences and to surface needs that a fixed questionnaire cannot capture. Recruitment will use multiple channels to reduce the risk of falling short of the target.

Platform Completion and Improvement

The core minimum viable product, including authentication and onboarding, session tracking, gradual reduction mode, the progress dashboard, and the optional AI chatbot, has been implemented as described in Chapter Three and validated in Chapter Four. Further improvement work focuses on robust local backups and cloud synchronisation so that updates never lose user progress, flexible currency and device pricing to address inaccurate financial calculators reported in competitor reviews, and a stable, empathetic conversational experience that does not fail with errors.

Usability Testing

With the minimum viable product now functional and technically validated, usability will be assessed through think-aloud testing with a small group of participants working through standardised task scenarios. The System Usability Scale will be administered afterwards, with

a score of seventy or above taken to indicate acceptable usability. Accessibility testing across assistive technologies will run alongside, and both will feed into iterative refinement before wider release.

Pilot Study and Evaluation

A single-arm feasibility study will then recruit UK-based adult daily vapers who are motivated to quit. It will capture baseline demographics and vaping history, automated engagement metrics, and outcomes at four-week follow-up, including abstinence and satisfaction. Quantitative data will be analysed using descriptive statistics, retention analysis, and correlation testing, while qualitative interviews, stratified by quit method, engagement level, and outcome, will be analysed thematically. A convergent design will integrate both strands to explain the engagement patterns observed.

Further Ideas to Improve the Platform

Beyond the planned work, the following ideas would strengthen the platform. Each is linked to evidence from this report or to the project's wider goals, and they are offered as candidates for prioritisation rather than firm commitments.

- **Just-in-time support at moments of craving:** using logged triggers and time-of-day patterns to deliver a coping prompt or breathing exercise at the moments a user is most at risk, rather than only on a fixed schedule.
- **Craving prediction from self-monitoring data:** applying a lightweight model to a user's own logged history to anticipate high-risk periods and pre-empt them, turning passive tracking into active support.
- **Adaptive reduction schedules:** adjusting the tapering plan in response to actual adherence, easing the pace after a difficult week rather than holding a fixed curve, which respects user autonomy and reduces the chance of giving up.
- **AI-assisted community moderation:** using the language model to flag toxic or harmful posts for review, directly addressing the weak moderation that undermines competitor community features.

- **Signposting to NHS services and nicotine replacement:** connecting users to local Stop Smoking Services and evidence-based nicotine replacement guidance, which supports the goal of NHS integration and provides a bridge to clinical help when needed.
- **A personalised health recovery timeline:** showing evidence-based improvements that occur at intervals after stopping, tailored to the user's history, to reinforce motivation.
- **Data export for clinicians:** allowing users to export their own progress to share with a GP or healthcare professional, supporting continuity of care and user agency over their data.
- **Equity features:** multi-language support and adjustable reading levels, so that the application reaches the disadvantaged populations that current paid products exclude.
- **A built-in comparison of quitting pathways:** offering both abrupt and gradual routes and measuring their relative effectiveness, which would directly address a gap in the literature where no such comparison exists for vaping.
- **Continuous in-app feedback:** brief, optional one-tap prompts inside the application, so that improvement is driven by ongoing user input rather than one-off surveys.

CHAPTER SIX - CONCLUSION

VapeFree responds to a clear and growing public health need with a product that is free, flexible, and built specifically for how people vape in the United Kingdom. The requirements phase has produced strong, convergent evidence from an analysis of 1,178 competitor reviews and an initial user survey, and that evidence has been translated into a concrete feature design and a working minimum viable product. The research pipeline that produced this evidence is fully implemented and has passed its automated tests, and the same governance techniques it demonstrates carry through to the application. The mobile platform has been implemented and subjected to technical validation, including sixty-eight automated unit and component tests (all passing), manual verification on Expo Go connected to Firebase emulators, and an optional cloud-built Android APK, as reported in Chapter Four. The next steps are to expand the survey, complete summative usability testing and accessibility evaluation, and conduct the pilot study set out in Chapter Five. This report will be updated as each of those evaluation phases is completed.

Limitations

Several limitations should be noted alongside these findings. The most important is the small size of the survey sample: only four responses ($n = 4$) were available at this stage, so the survey evidence is indicative rather than statistically representative and cannot be generalised to the wider UK vaping population. The sample is also demographically narrow, being confined to respondents aged 18 to 34, which leaves the needs of older vapers unexamined. More broadly, although a minimum viable product has been built and technically validated as described in Chapter Four, usability testing with participants and pilot outcome data are not yet available; those steps are future scope rather than part of this report. Conclusions about real-world engagement and cessation effectiveness therefore rest on formative requirements evidence and developer-led device testing rather than on independent participant evaluation. Finally, the

competitor analysis, although based on a large corpus of 1,178 reviews, draws exclusively on Google Play Store reviews and relies on rating-based sentiment and keyword-based theme extraction, which capture broad patterns but may under-represent the views of users who do not leave reviews and may miss nuance that finer-grained analysis would detect. These limitations are the direct motivation for the expanded recruitment and evaluation set out in Chapter Five.

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